

Before the FEDERAL COMMUNICATIONS COMMISSION Washington, D.C. 20554

In the Matter of Space Exploration Holdings, LLC Application for Orbital Data Center **ICFS File No. SAT-LOA-20260108-00016** Application for Launch and Operating Authority) **Call Sign: S00798** for the SpaceX Orbital Data Center System) **Public Notice: DA-26-113**

PETITIONER'S CONSOLIDATED MASTER BRIEF IN SUPPORT OF PETITION TO DENY

William Stewart (the Petitioner) hereby submits this Consolidated Master Brief in support of his active Petition to Deny. This brief synthesizes, codifies, and unifies the complete technical, environmental, mathematical, and financial objections presented across the Petitioner's initial filing, subsequent supplements, and formal technical rebuttals.

The purpose of this consolidated filing is to provide the Space Bureau with a single, comprehensive record of the critical safety, regulatory, and scientific failures inherent in the Applicant's proposal to operate a constellation of one million AI-processing satellites. The Commission cannot legally or responsibly issue a final public interest determination on this docket without systematically addressing every structural deficiency detailed herein.

I. Procedural History and Unlawful Exclusion of Physical Records

The Petitioner initiated this challenge on February 7, 2026, by executing a formal Petition to Deny during a documented Electronic Comment Filing System (ECFS) and ICFS portal outage. Operating under valid administrative protocols during system outages, the Petitioner delivered a comprehensive, notarized physical evidentiary package to the FCC Reference Information Center via tracked mail on February 17, 2026.

As documented in the Petitioner's formal Ex Parte record, the Space Bureau improperly refused to scan, process, or upload this physical filing into the electronic system, claiming the physical submission was irrelevant once the electronic portal returned to service. This administrative lockout created an artificial gap in the public record and constitutes a clear violation of basic due process. This electronic brief serves to permanently anchor those original physical arguments within the active electronic docket.

II. The Radiator Paradox and Thermodynamic Impossibilities

The core engineering architecture of the proposed Gen3 AI-processing satellites relies on a fundamental misrepresentation of thermodynamics. In its prior assertions, the Applicant claimed that high-density cluster operations are sustainable due to the natural 3-Kelvin cooling environment of Deep Space. This represents what the Petitioner terms the Radiator Paradox:

1. **The Vacuum Heat Trap:** While the cosmic microwave background sits at approximately 2.7 Kelvin, space is a vacuum. In a vacuum, heat cannot be dissipated via conduction or convection. It can only be rejected through thermal radiation.
2. **The Stefan-Boltzmann Constraint:** Heat rejection from an orbital asset is strictly governed by the Stefan-Boltzmann Law, where power radiated is a function of emissivity, the Stefan-Boltzmann constant, surface area, and absolute temperature ($P = e \cdot \sigma \cdot A \cdot T^4$). To operate server-grade AI-processing hardware at a density matching terrestrial data centers (approximately 15 to 20 kilowatts per cluster), each satellite must reject massive thermal loads.
3. **The Area Discrepancy:** To maintain standard operational temperatures for silicon processing components (approximately 350 Kelvin), each satellite requires a massive radiator surface area that far exceeds the physical dimensions disclosed in the Applicant's technical exhibits.

By completely omitting the mass, surface area, and resultant aerodynamic cross-section of these mandatory thermal management systems, the Applicant has submitted non-reproducible scientific data that violates the Information Quality Act standards for objectivity and transparency.

III. Collision Software Circumvention and the Solar Array Data Flaw

The Applicant's debris mitigation exhibits fail basic scientific scrutiny and rely on a critical data manipulation trick to pass the Commission's safety thresholds:

1. **Omission of Solar Array Mass and Area:** To satisfy the standard NASA Debris Assessment Software (DAS) limits for passive decay collision probabilities, the Applicant's models explicitly exclude the massive cross-sectional area of the satellite solar arrays, running calculations exclusively on the compact satellite bus. This structural omission artificially deflates the calculated debris generation risk on the public record.
2. **The Self-Certified Zero Metric:** In its May 29, 2026 clarifications, the Applicant flatly refused to provide standard, verifiable NASA DAS logs for small object collision risks. Instead, the Applicant relied on proprietary modeling via the European Space Agency DRAMA tool and simply declared its casualty risk to be considered zero based on unverified fragmentation assumptions. The Commission cannot accept commercial self-certification in place of transparent, reproducible public data logs.

IV. Orbital Hyper-Density and the 99,900-Satellite Group 1 Corridor

The Applicant's recent disclosure of its finalized six-altitude configuration confirms the Petitioner's warnings regarding predatory orbital space grabbing.

The Applicant seeks authorization to compress **99,900 satellites** into a hyper-narrow, 18-kilometer operational corridor spanning strictly from 550 kilometers to 568 kilometers in altitude. Forcing nearly 100,000 highly active assets into a narrow band creates an unprecedented localized traffic highway. This extreme spatial density exponentially increases the statistical probability of localized collision loops and cascade events, known as the Kessler Syndrome, which cannot be managed by autonomous maneuvering systems alone.

V. The Upper Low Earth Orbit Space-Junk Waiver Request

The most severe threat to orbital sustainability is found in Section 4(d) of the Applicant's recent supplement, where they openly request a sweeping, blanket waiver of the Commission's atmospheric reentry rules (47 CFR § 25.114) for all assets operating above 600 kilometers.

The Applicant frames this as a sustainability initiative, claiming they require flexibility to dump dead, non-operational satellites into high-altitude Earth disposal orbits or heliocentric deep-space tracks. In reality, this is an attempt to bypass the fuel mass and engineering costs required to safely deorbit heavy data processing units. Authorizing a commercial operator to permanently abandon dead, unguided hardware in the upper low Earth orbit environment turns a global commons into an unregulated corporate junkyard. At these altitudes, atmospheric drag is negligible, meaning these dead satellites represent permanent, multi-century collision hazards.

VI. Environmental Impacts: The Twilight Extension and Skyglow

The unprecedented volume of one million satellites invalidates the Commission's standard National Environmental Policy Act (NEPA) categorical exclusions, presenting extraordinary circumstances that demand a full Environmental Impact Study:

1. **The Twilight Extension Paradox:** Assets stationed at the upper limits of the requested operational envelope remain fully illuminated by the sun long after twilight on the ground. Lower satellites enter the Earth's shadow rapidly, but a fleet of this scale at higher altitudes will continuously reflect sunlight into the middle of the night.
2. **Cumulative Skyglow:** The reflection from one million processing surfaces will increase diffuse skyglow far beyond the 10 percent safety threshold established by the international astronomical community. This creates a permanent artificial haze that disrupts ground-based astronomy, hinders the detection of near-Earth asteroids, and fundamentally alters terrestrial night sky visibility.

VII. Corporate Acknowledgement of Financial Non-Viability

The hyper-accelerated regulatory path carved out for this application stands in stark opposition to the Applicant's own internal financial acknowledgements. In its Pre-IPO S-1 financial

disclosure statement executed on April 22, 2026, the Applicant explicitly warned prospective investors that the Orbital Data Center project faces significant technical complexity and may never achieve commercial viability.

The public interest is heavily disserved when the Commission rushes to grant unprecedented rule waivers, space-grabbing corridor monopolies, and data self-certifications for a project that the operator itself admits is a highly speculative risk.

VIII. Conclusion

The administrative record demonstrates that the SpaceX Orbital Data Center application is built upon thermodynamic contradictions, missing safety logs, intentionally skewed cross-sectional data, and an unsustainable reliance on rule waivers.

The Petitioner respectfully requests that the Space Bureau deny the application in its entirety, or, at a minimum, defer all launch and operational authorizations until the Applicant submits reproducible NASA DAS logs, includes solar array metrics in all safety calculations, and undergoes a full, independent Environmental Impact Study.

Respectfully submitted,

/s/ William Stewart

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Petitioner

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